5. A family of 3 people went out for dinner in a restaurant. The cost of two dosai, three idlies and two vadais is ₹ 150. The cost of the two dosai, two idlies and four vadais is ₹ 200. The cost of five dosai, four idlies and two vadais is ₹ 250. The family has ₹ 350 in hand and they ate 3 dosai and six idlies and six vadais. Will they be able to manage to pay the bill within the amount they had ?

1.4.3 (iii) Gaussian Elimination Method

This method can be applied even if the coefficient matrix is singular matrix and rectangular matrix. It is essentially the method of substitution which we have already seen. In this method, we transform the augmented matrix of the system of linear equations into row-echelon form and then by back-substitution, we get the solution.

Example 1.27

Solve the following system of linear equations, by Gaussian elimination method :

$4x+3y+6z=25,\ x+5y+7z=13,\ 2x+9y+z=1.$

Solution

Transforming the augmented matrix to echelon form, we get

$$\left[\begin{array}{ccc|c} 4 & 3 & 6 & 25 \\ 1 & 5 & 7 & 13 \\ 2 & 9 & 1 & 1 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 5 & 7 & 13 \\ 4 & 3 & 6 & 25 \\ 2 & 9 & 1 & 1 \end{array}\right] \xrightarrow{\substack{R_2 \to R_2 - 4R_1, \\ R_3 \to R_3 - 2R_1}} \left[\begin{array}{ccc|c} 1 & 5 & 7 & 13 \\ 0 & -17 & -22 & -27 \\ 0 & -1 & -13 & -25 \end{array}\right]$$

$$\xrightarrow{\substack{R_2 \to R_2 \div (-1), \\ R_3 \to R_3 \div (-1)}} \left[\begin{array}{ccc|c} 1 & 5 & 7 & 13 \\ 0 & 17 & 22 & 27 \\ 0 & 1 & 13 & 25 \end{array}\right] \xrightarrow{R_3 \to 17R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 5 & 7 & 13 \\ 0 & 17 & 22 & 27 \\ 0 & 0 & 199 & 398 \end{array}\right].$$

The equivalent system is written by using the echelon form:

$$x+5y+7z = 13, \quad \ldots (1)$$

$$17y+22z = 27, \quad \ldots (2)$$

$$199z = 398. \quad \ldots (3)$$

From (3), we get $z=\dfrac{398}{199}=2$.

Substituting $z=2$ in (2), we get $y=\dfrac{27-22\times 2}{17}=\dfrac{-17}{17}=-1$.

Substituting $z=2, y=-1$ in (1), we get $x=13-5\times(-1)-7\times 2=4$.

So, the solution is $(x=4, y=-1, z=2)$. ■

Note. The above method of going from the last equation to the first equation is called the **method of back substitution.**

Example 1.28



The upward speed $v(t)$ of a rocket at time t is approximated by $v(t)=at^2+bt+c,\ 0\le t\le 100$ where $a, b,$ and c are constants. It has been found that the speed at times $t=3, t=6$, and $t=9$ seconds are respectively, 64, 133, and 208 miles per second respectively. Find the speed at time $t=15$ seconds. (Use Gaussian elimination method.)

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